

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

SET THEORY

DIBY (DO IT BY YOURSELF)

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DIBY-01

State whether the following collections is a set or not?

- (i) The collection of natural numbers between 2 and 20
- (ii) The collection of numbers which satisfy the equation $x^2 - 5x + 6 = 0$
- (iii) The collection of prime numbers between 1 and 100.
- (iv) The collection of all intelligent women in Jalandhar:

Ans: (i) Yes, (ii) Yes, (iii) Yes, (iv) No

DIBY-02

Write the following set in builder form

- (i) Set of all rational number
- (ii) $\{2, 5, 10, 17, 26, 37, \dots \dots \dots\}$

Ans: (i) $\left\{x : x = \frac{p}{q}, p \in I, q \in N\right\}$
 (ii) $\{x : x = \lambda^2 + 1, \lambda \in N\}$

Consider set $A = \{1, 2, 3, 4, 5, 6, 7\}$ then:

- (i) number of subsets of set A
- (ii) of number of proper subsets of set A
- (iii) number of non-empty subsets of set A .
- (iv) number of non-empty proper subsets of set A .
- (v) number of non-empty subsets containing only odd numbers.
- (vi) number of subsets of A whose product of elements is even.
- (vii) number of subsets of A whose product of elements is odd.

Ans: (i) 128, (ii) 127, (iii) 127, (iv) 126, (v) 15, (vi) 112, (vii) 15

DIBY-04



Two finite sets have m and n elements respectively. The total number of subsets of first set is 992 more than the total number of subsets of the second set. Find the value of m & n .

Ans: $m = 10, n = 5$

DIBY-05

Value of $\left(\sqrt[6]{15 - 2\sqrt{56}} \cdot \sqrt[3]{\sqrt{7} + 2\sqrt{2}}\right)$ is

(A) 0

(B) 1

(C) -1

(D) $\sqrt{2}$

Ans: (B)

DIBY-06



Show that the square of $\frac{\sqrt{25 - 15\sqrt{3}}}{5\sqrt{2} - \sqrt{38 + 5\sqrt{3}}}$ is a rational number.

DIBY-07

Given the sets $A = \{1, 2, 3\}$, $B = \{3, 4\}$, $C = \{4, 5, 6\}$, then $A \cup (B \cap C)$ is

- (A) $\{3\}$ (B) $\{1, 2, 3, 4\}$ (C) $\{1, 2, 4, 5\}$ (D) $\{1, 2, 3, 4, 5, 6\}$

Ans: (B)

DIBY-08

There are 200 individuals with skin disorder, out of them 100 have been exposed to Chemical C_1 , 50 to Chemical C_2 , and 30 to both C_1 & C_2 . Find the number of individuals exposed to C_1 but not C_2 .

Ans: 70

Evaluate the unknowns in each of the following cases:

- (a)** $n(A \cup B), n(A \cap B)$, if $A \subset B, n(A) = 30, n(B) = 36$
 - (b)** $n(B)$, if $n(A) = 30, n(A \cup B) = 50, n(A \cap B) = 10$
 - (c)** $n(A \cup B)$, if $n(A) = 20, n(A' \cap B) = 8$
 - (d)** $n(A \cap B)$, if $n(A - B) = 30, n(B - A) = 16, n(A \cup B) = 60$
 - (e)** $n(A \cup B)$, if $n(A) = 58, n(B) = 102, n(A - B) = 38$
 - (f)** $n(A \cup B)$, if $n(S) = 70, n(A) = 40, n(B) = 50, n(A \cap \bar{B}) = 27$
- (Here S denotes the Universal Set)**

Ans:

- (a)** $n(A \cup B) = 36; n(A \cap B) = 30$
- (b)** $n(B) = 30$
- (c)** $n(A \cup B) = 28$
- (d)** $n(A \cap B) = 14$
- (e)** $n(A \cup B) = 140$
- (f)** **data is incorrect.**

DIBY-10



A survey shows that 61%, 46% and 29% of the people watched "3 idiots", "Rajneeti" and "Avatar" respectively. 25% people watched exactly two of the three movies and 3% watched none. What percentage of people watched all the three movies?

Ans: (7)

DIBY-11

A survey shows that 61%, 46% and 29% of the people watched "3 idiots", "Rajneeti" and "Avatar" respectively. 25% people watched exactly two of the three movies and 3% watched none. What percentage of people watched all the three movies?

There is a group of 265 person who like either singing or dancing or painting. In this group, 200 like singing, 110 like dancing and 55 like painting. If 60 persons like both singing and dancing, 30 like both singing and painting and 10 like all three activities, then the number of person who like only dancing and painting

- (A) 10 (B) 20 (C) 30 (D) 40**

Ans: (A)

DIBY-12

The set $(A \cap B')' \cup (B \cap C)$ is equal to

- (A) $A' \cup B \cup C$ (B) $A' \cup B$ (C) $A' \cap C'$ (D) $A' \cap B$

DIBY-13

Ans: (B)

Let X be the universal set for sets A and B , if $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n(A' \cap B')$ is equal to 300 provided $n(X)$ is equal to

- (A) 600 (B) 700 (C) 800 (D) 900

DIBY-14

Ans: (B)

In a group of 50 people, two tests were conducted, one for diabetes and one for blood pressure. 30 people were diagnosed with diabetes and 40 people were diagnosed with high blood pressure. What is the minimum number of people who were having diabetes and high blood pressure?

- (A) 0 (B) 10 (C) 20 (d) 30

Ans: (C)

DIBY-15

A class has 175 students. The following data shows the number of students obtaining one or more subjects. Mathematics 100, Physics 70, Chemistry 40; Mathematics and Physics 30, Mathematics and Chemistry 28, Physics and Chemistry 23; Mathematics, Physics and Chemistry 18. How many students have offered Mathematics alone?

- (A) 35 (B) 48 (C) 60 (D) 22

Ans: (C)

DIBY-16

In a class of 80 students numbered 1 to 80, all odd numbered students opt of Cricket, students whose numbers are divisible by 5 opt for Football and those whose numbers are divisible by 7 opt for Hockey. If the number of students who do not opt any of the three games is n , then $\frac{n}{4}$ is equal to

Ans: 7

DIBY-17

If $n(A \cap B) = 5$, $n(A \cap C) = 7$ and $n(A \cap B \cap C) = 3$, then the minimum possible value of $n(B \cap C)$ is

Ans: (3)

DIBY-18

Of the members of three athletic teams in a school 21 are in the cricket team, 26 are in the hockey team and 29 are in the football team. Among them, 14 play hockey and cricket, 15 play hockey and football, and 12 play football and cricket. Eight play all the three games. The total number of members in the three athletic teams is

- (A) 43 (B) 76 (C) 49 (D) None of these

Ans: (A)

DIBY-19

There are 70 students studying Math or English or German. 40 are studying Math, 30 studying German, 35 studying English and 15 in all three course. How many students are enrolled in exactly two of the courses?

- (A) 10 (B) 5 (C) 15 (D) 20

Ans: (B)

DIBY-20

From 50 students taking examinations in Mathematics, Physics and Chemistry, each of the student has passed in at least one of the subject, 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry and at most 20 Physics and Chemistry. What is the largest possible number that could have passed all three examination?

Ans: (14)

*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**